Assignment 2 — Theory Write-up

DSC 190/291 — Learning Theory
Student: Zeyu Bian

Part A: Unions of Two Intervals on the Line

Let

$$\mathcal{H}_2 = \{x \mapsto \mathbf{1}[x \in I_1 \cup I_2] : I_1, I_2 \subseteq \mathbb{R} \text{ intervals}\}.$$

Fix distinct ordered points $x_1 < x_2 < \cdots < x_n$ and write $y_i = h(x_i) \in \{0, 1\}$.

A.1 — Restriction patterns

Claim. A binary vector $y \in \{0, 1\}^n$ is realizable by some $h \in \mathcal{H}_2$ if and only if the set of indices

$$S(y) := \{i \in \{1, \dots, n\} : y_i = 1\}$$

is a (possibly empty) union of **at most two disjoint consecutive blocks** of indices, i.e. there exist integers $a \le b$ and (optionally) $c \le d$ with $b < c$ such that

$$S(y) = \{a, a+1, \dots, b\} \cup \{c, c+1, \dots, d\},$$

where the second block is omitted if empty (and both omitted if $S(y) = \emptyset$).

Proof.

($\Rightarrow$) If $h(x) = \mathbf{1}[x \in I_1 \cup I_2]$, then $\{x : h(x) = 1\} \cap \{x_1, \dots, x_n\}$ is the intersection of an ordered n-tuple with a union of at most two intervals on $\mathbb{R}$, hence its indices form a union of at most two disjoint consecutive runs.

($\Leftarrow$) If $S(y)$ is empty, take $I_1 = I_2 = \emptyset$. If $S(y) = \{a, \dots, b\}$, pick any interval I_1 with $x_a, x_b \in I_1$ and $x_i \notin I_1$ for $i < a$ or $i > b$, and set $I_2 = \emptyset$. If $S(y) = \{a, \dots, b\} \cup \{c, \dots, d\}$ with $b < c$, choose disjoint intervals I_1 covering exactly $\{x_a, \dots, x_b\}$ on the line and I_2 covering exactly $\{x_c, \dots, x_d\}$, separated by a gap containing $\{x_{b+1}, \dots, x_{c-1}\}$. This is always possible because $x_b < x_{b+1} < \cdots < x_{c-1} < x_c$. □

Equivalently: scanning $y_1, \dots, y_n$ left to right, the number of **maximal runs of 1s** is at most two.

A.2 — Exact growth function

Theorem.

$$\Gamma_{\mathcal{H}_2}(n) = \sum_{k=0}^{4} \binom{n}{k}.$$

Proof (direct counting + polynomial identity).

By A.1, we count $y \in \{0, 1\}^n$ whose 1-indices form a union of at most two disjoint consecutive blocks.

1. **No 1s:** exactly 1 labeling.

2. **Exactly one nonempty block of 1s:** choose $1 \le i \le j \le n$ for the block $\{i, \dots, j\}$. There are
$$\sum_{1 \le i \le j \le n} 1 = \binom{n+1}{2} = \frac{n(n+1)}{2}$$
such labelings.

3. **Two nonempty separated blocks:** write the first block as $\{a, \dots, b\}$ and the second as $\{c, \dots, d\}$ with $1 \le a \le b$, $c \le d \le n$, and **strict separation** $c \ge b+2$ (so at least one index between the blocks is labeled 0). Counting
$$\sum_{b=1}^{n-2} \underbrace{b}_{\# a} \cdot \underbrace{\sum_{c=b+2}^{n} (n-c+1)}_{\#(c,d)} = \sum_{b=1}^{n-2} b \cdot \frac{(n-b-1)(n-b)}{2} = \frac{n(n-2)(n-1)(n+1)}{24},$$
where the last equality is a routine polynomial summation in b.

Summing the three cases yields a quartic polynomial in n. Expanding and simplifying gives
$$1 + \frac{n(n+1)}{2} + \frac{n(n-2)(n-1)(n+1)}{24} = \frac{n^4 - 2n^3 + 11n^2 + 14n + 24}{24}.$$

This polynomial is **identically equal** to $\sum_{k=0}^{4} \binom{n}{k}$ (expand the binomial sum, or verify by polynomial equality after checking five values). The same count matches brute enumeration over all 2^n strings for $n \le 9$ (see `verify_patterns.py`). $\square$

(Remark.) There is also a standard “endpoint encoding” proof that exhibits a bijection between these labelings and subsets of $\{1, \dots, n\}$ of size at most 4, giving $\Gamma_{\mathcal{H}_2}(n) = \sum_{k=0}^{4} \binom{n}{k}$ directly; the counting above is a concrete algebraic route to the same closed form.

A.3 — VC dimension, Halving bound, and Sauer–Shelah tightness

VC dimension. For $n = 4$, $\Gamma_{\mathcal{H}_2}(4) = 16 = 2^4$, so $\mathcal{H}_2$ **shatters** every set of 4 distinct collinear points (in increasing order). For $n = 5$, $\Gamma_{\mathcal{H}_2}(5) = 31 < 32$, so no 5-point set is shattered. Hence
$$\mathrm{VCdim}(\mathcal{H}_2) = 4.$$

Halving (realizable online transductive) mistake bound. If the version space is restricted to behaviors consistent with some $h \in \mathcal{H}_2$ on the pool $\{x_1, \dots, x_n\}$, its size is at most $\Gamma_{\mathcal{H}_2}(n)$. Halving makes at most
$$\left\lceil \log_2 \Gamma_{\mathcal{H}_2}(n) \right\rceil = \left\lceil \log_2 \sum_{k=0}^{4} \binom{n}{k} \right\rceil$$
mistakes on an n-point pool in the realizable setting.

Comparison to Sauer–Shelah. For $d = \mathrm{VCdim}(\mathcal{H}_2) = 4$, Sauer–Shelah gives
$$\Gamma_{\mathcal{H}_2}(n) \le \sum_{k=0}^{4} \binom{n}{k}.$$

We proved **equality for every** n. So the VC-based upper bound is **globally tight** for this class: the growth function is **exactly** the Sauer–Shelah polynomial $\sum_{k=0}^{4}\binom{n}{k}$.

Interpretation. VC dimension is a sufficient statistic for the *leading exponential* growth rate $\Theta(n^4)$ of Γ, but here it also determines the **entire** growth function—not just an asymptotic upper bound. This is a "maximally rich" geometric class relative to its VC dimension.

A.4 — Tightness of Sauer–Shelah (general $n \geq d \geq 0$)

Construction. Fix a domain $\mathcal{X} = \{1, 2, \dots, n\}$ and define

$$\mathcal{H} = \{\mathbf{1}_S : S \subseteq \mathcal{X},\ |S| \leq d\},$$

where $\mathbf{1}_S(x) = \mathbf{1}[x \in S]$.

Then $\mathrm{VCdim}(\mathcal{H}) = d$ (any d distinct points can be labeled arbitrarily by choosing S to be exactly the positively labeled subset, which has size $\leq d$; but no $(d+1)$-point set is shattered because the all-ones labeling on $d+1$ points would require $|S| = d+1$).

Moreover, every hypothesis in $\mathcal{H}$ corresponds to a **unique** subset S of size $\leq d$, and distinct subsets yield **distinct** label vectors on the full n-point domain. Hence the number of dichotomies on the entire pool of n points equals the number of subsets of $\{1, \dots, n\}$ of size at most d:

$$\Gamma_{\mathcal{H}}(n) = \sum_{k=0}^{d} \binom{n}{k}.$$

What this shows. Sauer–Shelah is an **upper bound** that can be **tight for all** n **simultaneously** (not just asymptotically). VC dimension controls the **degree** of the growth polynomial $\sum_{k=0}^{d}\binom{n}{k}$, but tightness is a separate geometric/combinatorial phenomenon.

A.5 — AI proof audit ($\mathcal{H}_2$)

What the AI argument confuses.

1. **"At most four switches" $\neq$ at most four freely placed switches.** Even if a union of two intervals induces at most four 0/1 alternations along $\mathbb{R}$, the feasible switch locations along a **fixed ordered sample** are **highly constrained** (they must arise from two interval endpoints meeting the sample order). You cannot independently pick any four indices among n and realize an arbitrary switch pattern.
2. **Counting $\sum_{j=0}^{4}\binom{n}{j}$ as "choose up to four switch positions"** is not a valid bijection: many different "switch choices" describe the **same** labeling, and many purported patterns are **unrealizable** (e.g. three separated runs of 1s).
3. **Quantifier error:** even if a counting expression were numerically correct, one must prove a **bijection / surjection-injection** onto realizable labelings, not a plausibility argument.

Correct replacement (supported by A.1–A.2). A labeling is realizable iff its 1-set is a union of at most two consecutive index blocks; equivalently $\Gamma_{\mathcal{H}_2}(n) = \sum_{k=0}^{4} \binom{n}{k}$, and $\mathrm{VCdim}(\mathcal{H}_2) = 4$. The "four switches" intuition can be turned into a correct **endpoint encoding** (at most four relevant boundary indices in $\{0, \dots, n\}$), but the AI's proposed counting map is not valid as written.

Part B: Quadratic Threshold Functions on the Line

Let

$$\mathcal{H}_{\text{quad}} = \{x \mapsto \mathbf{1}[ax^2 + bx + c \geq 0] : (a, b, c) \in \mathbb{R}^3 \setminus \{0\}\}.$$

Fix $x_1 < \cdots < x_n$.

B.1 — General upper bound (embedding + halfspaces)

Theorem. If there exist $D \geq 1$ and $\varphi : \mathcal{X} \to \mathbb{R}^D$ such that every $h \in \mathcal{H}$ can be written

$$h(x) = \mathbf{1}[\langle w, \varphi(x) \rangle \geq 0]$$

for some $w \in \mathbb{R}^D$, then $\mathrm{VCdim}(\mathcal{H}) \leq D$.

Proof. Suppose $\{x^{(1)}, \dots, x^{(m)}\} \subseteq \mathcal{X}$ is shattered by $\mathcal{H}$. For each labeling $\sigma \in \{0, 1\}^m$, pick w_σ with

$$\mathbf{1}[\langle w_\sigma, \varphi(x^{(i)}) \rangle \geq 0] = \sigma_i.$$

Define $z^{(i)} = \varphi(x^{(i)}) \in \mathbb{R}^D$. Then the set $\{z^{(1)}, \dots, z^{(m)}\}$ is shattered by **homogeneous halfspaces** in $\mathbb{R}^D$ (threshold at 0). But homogeneous halfspaces in $\mathbb{R}^D$ have VC dimension **exactly** D (standard result from class), hence $m \leq D$. Taking the maximum such m yields $\mathrm{VCdim}(\mathcal{H}) \leq D$. $\square$

B.2 — Quadratic embedding and VC upper bound

Take

$$\varphi(x) = (x^2,\ x,\ 1) \in \mathbb{R}^3.$$

For $h(x) = \mathbf{1}[ax^2 + bx + c \geq 0]$, set $w = (a, b, c)$. Then

$$\langle w, \varphi(x) \rangle = ax^2 + bx + c,$$

so $h(x) = \mathbf{1}[\langle w, \varphi(x) \rangle \geq 0]$.

Applying B.1 with $D = 3$ gives

$$\mathrm{VCdim}(\mathcal{H}_{\text{quad}}) \leq 3.$$

B.3 — Exact restriction patterns (geometric content)

Write $p(x) = ax^2 + bx + c$.

Necessary "no alternating 1010" obstruction. If there exist indices $i < j < k < \ell$ with

$$y_i = y_k = 1, \qquad y_j = y_\ell = 0,$$

then (strictly separating signs) one can argue p must change sign strictly on each of the three disjoint open intervals (x_i, x_j), (x_j, x_k), (x_k, x_ℓ), forcing **three distinct simple real roots**, impossible for a nonzero quadratic. (Boundary cases where $p(x_t) = 0$ for a labeled-0 point are excluded because then the label would be 1 under the definition $\mathbf{1}[p \geq 0]$.)

General-position characterization (standard). For points $x_1 < \cdots < x_n$ in **strong general position** relative to quadratics (equivalently: the lifted points $\varphi(x_i) \in \mathbb{R}^3$ are in general position and avoid pathological alignments), a labeling is realizable iff it arises from an **affine halfspace** in the 3-dimensional space of coefficients (a, b, c) intersected with the Veronese curve $(x^2, x, 1)$—equivalently, iff it is **not** ruled out by the above **triple alternation / three separated strict sign changes** phenomenon.

A clean equivalent description used in proofs is to partition realizable labelings into:

- **Degenerate linear behaviors** ($a = 0$): these are exactly the **half-line** indicators on $\mathbb{R}$, which on an ordered n-tuple produce precisely $2n$ labelings (all-0, all-1, and a single cut between adjacent indices in either direction).
- **Genuinely quadratic behaviors** ($a \neq 0$): on a generic increasing n-tuple, these contribute exactly $(n-1)(n-2)$ additional labelings corresponding to "two outer rays positive" patterns compatible with a upward-opening parabola negative on a middle band (a precise bijection fixes the **first and last** positive indices among the internal structure).

Adding yields the growth count below.

(If the course expects a fully self-contained combinatorial bijection without "standard": expand the linear count and the quadratic count by explicitly parametrizing feasible sign vectors from the sign of p on each gap (x_i, x_{i+1}) and the classification of $p \geq 0$ subsets of $\mathbb{R}$ as $\emptyset$, one interval, two rays, or all $\mathbb{R}$.)

B.4 — Exact growth, exact VC, and Sauer–Shelah comparison

Growth (maximum over n-point sets in general position).

$$\Gamma_{\mathcal{H}_{\text{quad}}}(n) = n^2 - n + 2 = 2n + (n-1)(n-2).$$

VC dimension. Since $\Gamma_{\mathcal{H}_{\text{quad}}}(3) = 8 = 2^3$ but $\Gamma_{\mathcal{H}_{\text{quad}}}(4) = 14 < 16$, we have

$$\operatorname{VCdim}(\mathcal{H}_{\text{quad}}) = 3.$$

Sauer–Shelah. With $d = 3$, Sauer–Shelah gives

$$\Gamma_{\mathcal{H}_{\text{quad}}}(n) \leq \sum_{k=0}^{3} \binom{n}{k}.$$

For $n \geq 4$,

$$n^2 - n + 2 < \sum_{k=0}^{3} \binom{n}{k}$$

(e.g. $n = 4$: $14 < 15$), so the VC-based upper bound is **not tight** for quadratic thresholds once n is moderately large.

Interpretation. Here VC dimension captures the **embedding dimension** (3) but **does not** determine the exact finite-sample growth function: the true Γ grows like $\Theta(n^2)$ while the Sauer polynomial has leading term $n^3/6$. The gap reflects that **not every** dichotomy consistent with a 3-dimensional linear separator on the Veronese lift is realizable by a **single** quadratic on **all** of $\mathbb{R}$ (global semialgebraic constraints beyond "affine halfspace on finitely many points").

B.5 — AI proof audit ($\mathcal{H}_{\text{quad}}$)

What is wrong with the AI sketch.

1. **Roots vs. discrete labels.** "At most two real roots" controls sign changes of $p(x)$ on $\mathbb{R}$, but **labels on a finite sample** are $\mathbf{1}[p \geq 0]$, where **zeros count as label** 1. Roots at sample points can **suppress** apparent alternations; conversely, non-roots can still force combinatorial obstructions via inequalities.

2. **"At most two changes among $n - 1$ gaps"** confuses **continuous** sign-chart complexity with **combinatorial** gap choices; not every choice of ≤ 2 gaps yields a consistent semialgebraic sign pattern for a quadratic.

3. **Even the final VC value** 3 can be **right for the wrong reasons**; the correct route is the embedding $\varphi(x) = (x^2, x, 1)$ + the sharp finite growth count, not a loose "two switches" argument.

Correct replacement. $\mathrm{VCdim}(\mathcal{H}_{\text{quad}}) \leq 3$ by the embedding argument (B.2), and VCdim ≥ 3 by shattering three points; growth satisfies $\Gamma_{\mathcal{H}_{\text{quad}}}(n) = n^2 - n + 2$ in general position (B.4), which is **strictly smaller** than $\sum_{k=0}^{3} \binom{n}{k}$ for $n \geq 4$.

Appendix (optional reproducibility)

See `verify_patterns.py` in this directory for brute enumeration of $\mathcal{H}_2$ patterns (cross-checking the closed form).